David Dobáš

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Summary

Robotics engineer with a foundation in mathematical physics and complex systems, working on simulation and robot learning for navigation, manipulation, and humanoid whole-body control.

At Innate, built the production MuJoCo-based simulator for innate-os and train navigation and manipulation policies; previously trained and deployed RL and imitation policies on Unitree G1 humanoids (BeyondMimic, Sonic, BFM-Zero, OmniXtreme).

Proven at bridging sim-to-real, from large-scale live humanoid performances (8,000+ audience) to shipping a developer-facing robot simulator.

Professional Experience

Member of Technical Staff *Palo Alto, USA*
Innate *June 2026 – present*

- Built the production innate-os simulator, a MuJoCo digital twin of the robot that streams live world state to a Three.js 3D view in the browser, replacing the earlier Genesis-based simulator.
- Shipped the simulator to developers: one-command installer (macOS/Linux/WSL), multi-arch Docker images via CI, usage telemetry, and a hosted per-visitor browser demo.
- Modeled grasping physics (gripper contact, precise collision geometry, props) and built challenge scenarios with ground-truth judging for navigation and manipulation.
- Train navigation and manipulation policies; integrated the learned nav policy on the robot and built a benchmark with PointNav, ObjectNav and VLN scenarios.
- Tuned the classical navigation stack (Nav2/MPPI acceleration limits, lidar-consistent mapping for AMCL) and built the data-collection pipeline: H.264 dataset encoding, episode replay, live training progress.

Robotics Research Engineer *SF, USA / KL, Malaysia*
Ultimate Bots *Jan 2026 – May 2026*

- Deployed and fine-tuned latest humanoid foundation models (NVIDIA Sonic, BFM-Zero, OmniXtreme) on Unitree G1, adapting policies to hardware dynamics and control constraints.
- Trained PPO-based motion imitation policies using BeyondMimic in mjlab/MuJoCo from mocap/video, with custom domain randomization for robust sim-to-real transfer.
- Led deployment and real-time debugging of whole-body policies in large-scale live dance performance (8,000+ audience, 2.7M+ views), fixing instability, contact dynamics, URDF mismatches, and hardware variability via iterative optimization.
- Developed locomotion-imitation policy blending for stable coordinated behaviors.
- Built GMR-based video retargeting pipelines with stability post-processing; scaled training via Nebius cloud.
- Taught humanoid RL workshops (UC Berkeley, Nebius Build SF) on Isaac Lab, MuJoCo, retargeting, imitation, and real-robot deployment.

Founding AI Engineer *Paris, France*
Gigi *July 2025 – Aug 2025*

- Architected the core AI infrastructure for a people-search engine, enabling natural-language retrieval across tens of thousands of profiles.
- Engineered a compositional query generation engine (Natural Language to SQL) and a custom trait-extraction pipeline.
- Designed a proprietary Network & Reputation scoring algorithm using iterative graph propagation, serving as the fundamental ranking logic for the platform.

Quantitative Trader & Researcher *Prague, Czech Republic*
RSJ Securities a.s. *Feb 2024 – Mar 2025*

- Managed real-time execution of high-frequency trading algorithms as a Market Maker. Directly responsible for strategic parameter adjustment to maintain profitability and liquidity during volatile market conditions.

Selected Projects

TED2026 Talk: How to teach a robot to dance
"All of Us" session · Vancouver, Apr 2026

- Spoke on how advances in AI have transformed the way humanoid robots learn, from thousands of simulations to mimicking human movement in minutes.
- Ran a live demonstration with technologist Bilawal Sidhu: two humanoids mirrored human movement in real time via direct body control, then performed a dance learned in a single afternoon.

Large-Scale Humanoid Performance Deployment

- Led end-to-end deployment of coordinated whole-body dance routines on 8 Unitree G1 humanoids for DJ Wukong Chinese New Year live show (8,000+ audience, 2.7M+ online views).
- Built complete pipeline: mocap + GMR retargeting with heavy post-processing (drift/symmetry/smoothing correction) → BeyondMimic + PPO training in mjlab with custom domain randomization → locomotion + imitation policy blending.
- Resolved real-world challenges including balance instability, outdated URDF mismatches, open-loop control constraints, hardware variability, and minimal rehearsal time through rapid calibration and debugging.

Humanoid Motion Learning from Video

- Built pipeline for extracting motion from video → retargeting → RL training in simulation.
- Enabled scalable generation of training data without motion capture.

Humanoid Training Workshops (UC Berkeley, Nebius Build SF)

- Led workshops on humanoid RL pipelines (Isaac Lab, MuJoCo, mjlab).
- Taught motion retargeting, imitation learning, and sim-to-real deployment strategies.
- Supported teams deploying policies in simulation and on real robots.

Research and Publications

q-Exponential Random Graphs: higher-order networks from simple constraints
D. Dobáš, D. Garlaschelli, P. Jizba · *arXiv, May 2026*

- Generalized Exponential Random Graphs beyond Shannon entropy by maximizing Uffink entropies, yielding q-exponential network ensembles with dependent edges.
- Extended the Erdős-Rényi and configuration models under the same constraints (link count, degree sequence), revealing a sparse-to-dense phase transition absent in the classical models.
- Showed that richer assortativity and clustering emerge, letting link sparsity and triadic closure coexist: higher-order structure from simple constraints.

Scale-Invariant Network Reconstruction Model *Sept 2023 – Sept 2024*

- Conducted research at Leiden University (Econophysics & Network Theory Group) applying scale-invariant models to reconstruct interbank loan networks, improving systemic risk estimation methods.

Education

Czech Technical University in Prague *Prague, Czech Republic*
MSc (Graduated with Honors) in Mathematical Physics *Sept 2022 – June 2025*

- GPA: 3.94/4.0
- Thesis: *Complex networks modelling using non-Shannonian entropies* (Basis for arXiv:2605.31209).

prg.ai *Prague, Czech Republic*
Selective Minor in Machine Learning & AI *May 2021 – present*

- Selected for elite inter-university AI program (GPA: 4.0/4.0).
- Deep Reinforcement Learning course: full AlphaZero reimplementation from scratch in PyTorch.

Leiden University *Leiden, Netherlands*
Visiting Graduate Researcher in Physics *Sept 2023 – Jan 2024*
- Research Focus: Complex network reconstruction under Prof. Diego Garlaschelli.

KAIST *Daejeon, South Korea*
Visiting Graduate Researcher in Physics *Feb 2023 – June 2023*
- Research Focus: Statistical Physics of Complex Systems under Prof. Hawoong Jeong.

Czech Technical University in Prague *Prague, Czech Republic*
BSc (Graduated with Honors) in Mathematical Engineering *Sept 2019 – June 2023*
- GPA: 3.97/4.0

Technical Skills

ML Frameworks & Tools: PyTorch (AlphaZero reimplementation, transformers, CNNs etc.), MuJoCo/mjlab, Isaac Lab, ROS, Nebius Cloud, Git, Python (Advanced)

Robotics & AI: Deep RL (PPO), Imitation Learning, Humanoid Foundation Models (Sonic, BFM-Zero, OmniXtreme), Domain Randomization, Sim-to-Real, Motion Retargeting, Whole-Body Control

Core Math & Physics: Stochastic Differential Equations, Linear Algebra, Probability & Statistics, Nonequilibrium Statistical Physics

Languages

Czech: Native

English: C1 (Cambridge Advanced)

Korean: A2